

Lebedev Finite-Difference Maxwell Solver

Code Overview, Parameters, and the DDH03 Borehole Example

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Part I: Code Description

1. Overview

This code implements the Lebedev finite-difference (FD) scheme for the time-harmonic (frequency-domain) Maxwell equations in 3D, following Davydcheva, Druskin, and Habashy (2003), hereafter DDH03. The solver targets borehole electromagnetic logging simulations: a magnetic dipole source on the borehole axis, surrounded by cylindrical layers and possibly intersected by a dipping planar geological boundary.

The Lebedev scheme partitions the grid into two interleaved subgrids (P-nodes and R-nodes, based on the parity of the index sum $i+j+k$). Electric-field components live at R-nodes, magnetic-field components at P-nodes. Four staggered "clusters" (C000, C101, C110, C011) tile the field components across these subgrids, generalizing the Yee grid to handle fully anisotropic conductivity tensors without unphysical cross-coupling.

Code modules:

- `grid.py` — Grid construction (LebedevGrid3D, optimal geometric grids, hybrid axial grids)
- `media.py` — Conductivity assignment and interface averaging (EMMedia, nodal homogenization)
- `operators.py` — FD curl operators and boundary condition application
- `solver.py` — System assembly and linear solve
- `postprocess.py` — B-field extraction, multi-cluster averaging, axis sampling

2. Grid

2.1 Transverse grid (x and y)

Primary node spacings grow geometrically from a minimum step $h_{\min}$:

$$h_i = h_{\min} \cdot \alpha^{(i-1)}, \quad \alpha = \exp(\gamma\pi/\sqrt{k}), \quad \gamma = 1/\sqrt{2} \quad (\text{EM optimal})$$

Dual nodes are interleaved using spacings $\hat{h}_1 = h_{\min}/(1+\sqrt{\alpha})$ and $\hat{h}_i = h_i/\sqrt{\alpha}$ for $i \geq 2$. The full symmetric grid is formed by mirroring the positive half-axis, giving $M_x = M_y = 4k$ intervals. The

optimal-grid theory underlying this construction — including the choice $\gamma = 1/\sqrt{2}$ for EM diffusion — originates with Druskin and Knizhnerman (1999) and Ingerman, Druskin, and Knizhnerman (2000).

Parameter	Typical value	Description
$h_{\min}$	0.10–0.17 m	Minimum primary spacing (inner resolution)
k	3–6	Primary steps per half-axis; $M_x = 4k$. Controls both resolution and domain size.
L	300 m	Target domain half-length (k is auto-chosen to reach L if not specified)
γ (gamma)	$1/\sqrt{2}$	Progression exponent; $1/\sqrt{2}$ is EM-optimal per DDH03

Maximum transverse extent: $x_{\max} = h_{\min} \cdot \alpha^{(k-1)}$. With $h_{\min} = 0.10$ m: $k=3$ gives $x_{\max} \approx 1.3$ m, $k=4 \approx 2.8$ m, $k=5 \approx 5.3$ m, $k=6 \approx 9.3$ m. A minimum of roughly 0.7 skin depths of transverse coverage is needed for accurate results.

Grid alignment: If $h_{\min}$ is chosen as $R_{\text{interface}}/\alpha$ (the "magic" $h_{\min}$), the interface lands exactly on a primary node, eliminating interface averaging at that boundary. For all other $h_{\min}$ values, nodal homogenization (Section 4) handles the straddling cells.

2.2 Axial grid (z)

A hybrid grid: an equidistant inner zone $[z_{\text{tool_min}}, z_{\text{tool_max}}]$ resolves the tool uniformly; outside this range an optimal geometric expansion pushes the computational boundary far away without excessive nodes.

Parameter	Typical value	Description
$z_{\text{tool_min}}$	−3.5 m	Lower boundary of equidistant inner zone
$z_{\text{tool_max}}$	+2.5 m	Upper boundary of equidistant inner zone
n_{inner}	98	Number of equidistant steps in the inner zone (must be even)
k_{outer}	8	Optimal-geometric steps on each outer side

Total z-nodes: $4 \cdot k_{\text{outer}} + n_{\text{inner}} + 1$. With the values above, $M_z = 130$.

3. Conductivity Model

3.1 Supported geometry

The code supports a borehole logging geometry with three nested regions:

- **Cylindrical borehole:** radius R_BORE , isotropic conductivity σ_BORE . Occupies $r < R_BORE$.
- **Cylindrical invasion zone:** outer radius R_INV , isotropic σ_INV . Occupies $R_BORE < r < R_INV$.
- **Formation:** $r > R_INV$. May be divided into two half-spaces by a dipping planar boundary (normal N_HAT , offset D_PLANE). Each half-space can be independently isotropic or anisotropic.

The plane is $N_HAT \cdot x = D_PLANE$. $N_HAT = (\sin\theta, 0, \cos\theta)$ for a boundary dipping at angle θ from the borehole axis. D_PLANE sets the depth at which the plane crosses the axis.

3.2 Conductivity parameters

Parameter	Units	Description
SIGMA_BORE	S/m	Borehole fluid conductivity (isotropic)
SIGMA_INV	S/m	Invasion zone conductivity (isotropic)
SIGMA_ISO	S/m	Isotropic formation conductivity (one side of dipping plane)
SIGMA_T	S/m	Transverse conductivity of anisotropic formation (perpendicular to N_HAT)
SIGMA_N	S/m	Normal conductivity of anisotropic formation (along N_HAT)
R_BORE	m	Borehole radius
R_INV	m	Outer radius of invasion zone
DIP_RAD	rad	Dip angle of formation boundary from the borehole axis
D_PLANE	m	Signed offset of dipping plane along N_HAT from origin

The anisotropic formation tensor is the standard transverse isotropic (TI) form:

$$\sigma_ANISO = \sigma_T \cdot I + (\sigma_N - \sigma_T) \cdot N_HAT \otimes N_HAT$$

Conductivity along N_HAT is σ_N ; transverse to it is σ_T . Setting $\sigma_N = \sigma_T$ gives an isotropic medium.

4. Interface Averaging

Because the Lebedev grid is coarsely spaced, physical interfaces will generally not coincide with grid nodes. The conductivity assigned to a node whose dual cell straddles an interface directly affects accuracy. Two levels of treatment are used depending on how many interfaces a cell straddles.

4.1 Nodal homogenization for single-interface cells

For any node whose dual cell crosses exactly one interface, the nodal homogenization formula of Moskow et al. (1999) is applied. This computes an effective conductivity tensor by matching the continuous and discrete electromagnetic energy inner products over the dual cell.

Procedure for an interface with unit normal $\hat{n}$ between media σ_1 and σ_2 :

- Compute volume fraction f_{vol} of medium 1 in the dual cell (analytic integration over cell half-widths).
- Compute per-axis line fractions $f_{\text{line}}[\alpha]$: fraction of the grid edge in each direction passing through medium 1.
- Build the modified basis $\tilde{L} = [\hat{m} - \rho_m \odot \hat{n} \mid \hat{q} - \rho_q \odot \hat{n} \mid D_1 \odot \hat{n}]$, where $\hat{m}$, $\hat{q}$ are tangentials to $\hat{n}$, ρ encodes off-diagonal coupling for anisotropic media, and D_1 encodes per-axis line-average resistivity.
- Assemble energy matrix G from Schur complements of σ_1 and σ_2 .
- Solve $\Sigma_D = \tilde{L}^{-T} G \tilde{L}^{-1}$ for the effective nodal tensor.

For axis-aligned interfaces this reduces to the classical result: tangential components take the arithmetic mean $\bar{\sigma} = f_1 \sigma_1 + f_2 \sigma_2$, and the normal component takes the harmonic mean $\bar{\sigma} = 1/(f_1/\sigma_1 + f_2/\sigma_2)$. Tilted or anisotropic interfaces require the full tensor formula and introduce off-diagonal entries.

This nodal homogenization is applied at every single-straddling cell: at the borehole wall (radial normal $\hat{n}_{\text{r}}$), at the invasion boundary (radial normal $\hat{n}_{\text{r}}$), and at the dipping formation plane (constant normal N_{HAT}).

Two computational paths are supported depending on how the interface geometry is provided.

Exact geometry path (from `geometry_func`). When the interface is described by an explicit geometric object (cylinder, plane, level set), $\hat{n}$ is available analytically at each node. Volume and line fractions are computed by exact integration and the nodal formula is applied directly. This path is used for the borehole wall, invasion boundary, and dipping-plane interfaces in the DDH03 example.

Lookup-function path (from `sigma_func`). When conductivity is given only as a black-box callable, $\hat{n}$ is estimated from the local variation of σ within the dual cell. The code evaluates σ on a fine isotropic sub-grid, binarises the values, and identifies voxel pairs that straddle the interface. A plane is fitted through the physical midpoints of those crossing pairs by SVD; the last right singular vector (minimum-variance direction) gives $\hat{n}$. A planarity ratio $r = s_3/s_1$ measures confidence: if $r \geq 0.7$ the block has no clear interface direction and falls back to axis-aligned harmonic/arithmic averaging. For tensor-valued callables the SVD operates on $\text{tr}(\sigma)/3$ as a scalar proxy; the full tensor is used for the homogenization integrals once $\hat{n}$ is established.

4.2 Sequential nodal homogenization for doubly-straddled cells

When the invasion zone boundary (cylindrical, $r = R_{\text{INV}}$) and the dipping formation plane intersect near a cell, that cell straddles two interfaces simultaneously. Applying the nodal formula for only the cylindrical interface while using the pointwise outer medium is inconsistent, because the outer medium is itself inhomogeneous within the cell.

Sequential nodal homogenization resolves this with a two-step procedure, applying the nodal formula at each step:

- **Step 1 — Outer nodal homogenization:** Compute volume fractions f_{ani} and f_{iso} of the two outer-formation regions within the cell. Apply the nodal formula (Section 4.1) to these two materials with normal N_{HAT} , yielding an effective outer-medium tensor $\Sigma_{\text{outer}} = \tilde{L}^{\text{TT}1} G_{\text{out}} \tilde{L}^{\text{i}1}$. The inner material is identified as the one with the median conductivity value ($\text{tr}(\sigma)/3$), so the outer pair is always the lowest and highest conductivity materials in the cell.
- **Step 2 — Nodal homogenization:** Apply the single-interface formula (Section 4.1) between σ_{INV} and σ_{outer} along the cylindrical normal n_{hat_r} .

For cells straddling only the cylindrical boundary, only step 2 applies with the pointwise outer medium. For cells straddling only the planar boundary (outside R_{INV}), only the single-interface nodal formula with N_{HAT} applies. Sequential nodal homogenization degenerates gracefully to the standard single-interface treatment everywhere except at the small number of doubly-straddled cells near the intersection of the two boundaries. If the intermediate Σ_{outer} has a maximum eigenvalue exceeding three times the largest material conductivity — a symptom of near-degeneracy — the result is discarded and the cell falls back to the single-interface nodal formula.

5. Boundary Conditions

All outer faces use the DDH03 "combined" boundary condition (DDH03 eq. 6): clusters whose E-components are tangential to a face satisfy PEC ($E \times \hat{n} = 0$); clusters whose E-components are normal satisfy PMC. In practice all E-field DOFs at boundary R-nodes are set to zero (Dirichlet). Reflections are negligible when the boundary is at several skin depths from the source.

6. Linear System and Solve

The solver assembles and solves:

$$(C^T \mu^{-1} C - i\omega \sigma) E = b$$

C is the FD curl operator, $\sigma = \sigma + i\omega\epsilon$ is the complex conductivity ($\epsilon = \epsilon_0$ by default), and b is the magnetic dipole right-hand side. The system has $3 \cdot N_{\text{R}}$ unknowns. The default solver is `scipy.sparse.linalg.spsolve` (direct sparse LU). For large grids or benchmarking runs, `LGMRES` (`scipy.sparse.linalg.lgmres`) with Jacobi preconditioning is used instead, with tolerance 10^{-8} and up to 300 iterations.

k	h_min (m)	N_R	DOFs	Approx. solve time
3	0.166	11,069	33,207	~5 s
4	0.10	18,085	54,255	~10 s
5	0.10	28,885	86,655	~20 s
6	0.10	40,937	122,811	~60 s

7. Source and Multi-Cluster Field Extraction

7.1 Multi-cluster source

The source is a magnetic dipole at the origin. All four Lebedev clusters must be excited coherently: `build_rhs_multicl` distributes the dipole right-hand side across each cluster's native sub-grid nodes by trilinear interpolation. The source frequency ω , orientation (e.g., x-directed H_x), and position are all configurable.

7.2 Multi-cluster B-field averaging

After solving, B is computed from Faraday's law: $B = (1/i\omega) \nabla \times E$. To extract the physically correct field on the borehole axis, `extract_B_on_axis_multicl` performs a proper Lebedev inter-cluster average: for each target z , all four clusters contribute via trilinear interpolation from their respective native sub-grids to the target point, and the four values are averaged.

This inter-cluster averaging is essential. Reading only the on-axis P-nodes without interpolating the other clusters systematically biases the result by whichever sub-grid those nodes happen to belong to.

Part II: The DDH03 Benchmark Example

8. Model Geometry

The benchmark reproduces Figure 7 of DDH03: imaginary parts of the x-directed magnetic field ($\text{Im}(B_{xx})$ and $\text{Im}(B_{xz})$) on the borehole axis as the tool traverses a 60° -dipping anisotropic formation. The geometry is a borehole ($\sigma = 0.05$ S/m, $R = 0.1$ m) surrounded by a mud-filtrate invasion zone ($\sigma = 0.1$ S/m, $R = 0.6$ m). The outer formation is divided by a 60° dipping plane into an isotropic region ($\sigma = 0.5$ S/m, above the plane on axis) and a transversely isotropic region ($\sigma_T = 0.1$ S/m, $\sigma_N = 0.01$ S/m, below the plane on axis).

Parameter	Value	Description
f	52,650 Hz	Operating frequency
SIGMA_BORE	0.05 S/m	Borehole fluid
SIGMA_INV	0.10 S/m	Invasion zone
SIGMA_ISO	0.50 S/m	Isotropic formation (above dipping plane on axis)
SIGMA_T	0.10 S/m	Transverse conductivity of anisotropic formation
SIGMA_N	0.01 S/m	Normal conductivity (10:1 resistivity anisotropy)
R_BORE	0.10 m	Borehole radius
R_INV	0.60 m	Invasion zone outer radius
DIP_RAD	$\pi/3 = 1.0472$ rad (60°)	Formation dip angle from borehole axis
D_PLANE	-0.25 m	Plane crosses borehole axis at $z = -0.5$ m

The dipping plane crosses the axis at $z = D_PLANE/\cos(60^\circ) = -0.5$ m. The skin depth in the isotropic formation is $\delta \approx 3.1$ m at 52.65 kHz and 0.5 S/m.

9. Observables and the Crossing Diagnostic

Both $\text{Im}(B_{xx})$ (x-component, sensitive to transverse conductivity) and $\text{Im}(B_{xz})$ (z-component, sensitive to dip and anisotropy contrast) are always positive and decay away from the source. As the tool descends below the dipping plane, $\text{Im}(B_{xz})$ grows relative to $\text{Im}(B_{xx})$, and they cross at a depth z_{cross} .

The crossing depth z_{cross} where $\text{Im}(B_{xx}) = \text{Im}(B_{xz})$ is scale-invariant: independent of source moment, depending only on geometry and conductivity contrasts. It is the most robust single number for comparison with DDH03.

10. Interface Geometry in This Example

Three interface types are present simultaneously, making this a demanding test case for interface averaging:

- Borehole wall: cylindrical at $r = 0.10$ m, radial normal, isotropic-to-isotropic.
- Invasion boundary: cylindrical at $r = 0.60$ m, radial normal, isotropic-to-anisotropic (or isotropic-to-isotropic if the cell is also near the dipping plane).
- Dipping formation plane: constant normal $N_HAT = (\sin 60^\circ, 0, \cos 60^\circ)$, anisotropic-to-isotropic, tilted at 60° so the plane cuts through cells obliquely.

The dipping plane intersects the invasion boundary along a curve at $r \approx R_INV$ near $z = -0.5$ m. Cells in this vicinity straddle both interfaces simultaneously (doubly-straddled cells): roughly 550–650 cells depending on k , out of 11,000–41,000 total.

For all other interface cells, standard single-interface nodal homogenization (Section 4.1) applies. For doubly-straddled cells, sequential nodal homogenization (Section 4.2) applies: the two outer-formation regions are first collapsed into a single effective tensor via the nodal formula, which is then used in a second nodal homogenization at the cylindrical boundary. The spread in z_cross between pointwise outer-medium treatment and sequential nodal homogenization is 10–15 mm, a direct measure of sensitivity to the doubly-straddled cell treatment.

11. Convergence Results

With strategy E and appropriate h_min , the crossing depth converges to $z_cross \approx -1.29$ to -1.33 m across $k = 3$ –6:

k	h_min (m)	x_max	z_cross strat E	vs. DDH03
3	0.166 (magic)	2.16 m (0.70 δ)	–1.324 m	edge of band
4	0.10	2.80 m (0.90 δ)	–1.290 m	within band
5	0.10	5.32 m (1.72 δ)	–1.292 m	within band
6	0.10	9.32 m (3.01 δ)	–1.306 m	within band
DDH03	—	—	–1.29 to –1.33 m	reference

Note on $k=3$: With $h_min = 0.10$ m, $x_max = 1.30$ m (0.42 δ) and R_INV falls at 62% invasion fraction inside a cell. Both problems shift the crossing to $z \approx -1.72$ m. Setting $h_min = R_INV/\alpha = 0.166$ m places R_INV exactly on a primary node and raises x_max to 2.16 m (0.70 δ), restoring agreement. For $k \geq 4$, $h_min = 0.10$ m is sufficient because R_INV already falls at 72–96% of the way toward the next node.

References

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